Omer Rehan 24k-0760 BSCS-4K

Q1 (1): Following are issues faced by Cyber Nova through file based system.

- Data redundancy :- In different files some data stored multiple times
- Data Inconsistency: Different records across files on offices.
- Data Isolation: Data stored in different files. It becomes more difficult to understand
- Limited data sharing:- Not allowed any access between files or branches.
- Poor security control. No! Data protection and weak access of data.

(2): Centralized database helps in single integrated system for all branches.

It reduces duplication or data redundancy.

It implies data validation rules.

It permits role based access control and authentication

Efficient querying and indexing.

(3). By real time access employee can access data updating instantly. Access from any directory or location. Shared incidents logs and services records. Immediate access to accurate information.

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```
(1): Select * FROM CIENTS
     WHERE Country = 'Pakistan';

(2) select service-name, service-cost
    FROM services
    WHERE service-cost > 500;

(3) SELECT * FROM service-records
    WHERE Status = 'Completed';

(4) SELECT client-name
    FROM Clients
    WHERE Country = 'Pakistan';
(5) SELECT * FROM services
    WHERE service-cost BETWEEN 300 and 700;
(6) SELECT DISTINCT Locatation
    FROM offices;
(7) SELECT * FROM service-records
    ORDER BY service-date DESC;
(8) SELECT UPPER(Client-name)
    FROM Clients;
(9) SELECT Count (*) As tot-service-record
    FROM service-records;
(10) SELECT service-name, service-cost
     FROM service-records
     ORDER BY service-cost ASC;
```


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Q3 (1): INSERT INTO Borrow_records(Member-ID, Book_ID, borrow_date)
VALUES(104, 1, 2026-02-05);

(2). UPDATE Books SET Copies-available=-1
WHERE book_id=1;

(3) DELETE FROM MEMBERS
WHERE member-id=101;

(4) INSERT INTO MEMBERS(member_id, member_name, email, member_type)
VALUES(107, NULL, 'faizan@email.com', 'Premium');

(5): UPDATE BORROW-RECORDS SET book-id=5
WHERE record-id=1002;

(6): INSERT INTO Books (Title, author, copies-available, book_id)
VALUES('Network security', 'Ali Raza', 2, 4);

(7) UPDATE Borrow-records SET member-id=102
WHERE record_id=1003;

(8) DELETE FROM Books
WHERE book_id=3;

(9) INSERT INTO Borrow-records(record_id, member-id, book_id,
borrow-date, return-date)
VALUES(1005, 102, 3, '2026-02-07', '2026-02-15');

(10) ALTER TABLE Book
ADD Constraint Check-copies
CHECK (Check-copies >= 1);

Q4

1) There are 7 entities

- Book
- Book-Authors
- Book-copies
- Book_LOANS
- LIBRARY-BRANCH
- BORROWER
- PUBLISHER

2) Book = bookid

Book.AuthoRs = (Bookid, AuthorName) composite PK

PUBLISHER = Name

Book.COPIES = (Bookid, Branchid)

Book-LoANS = (Bookid, Branchid, CardNo)

LIBRARY_BRANCH = Branchid

BORROWER = CardNo

(3) Book = Publisher-name; each book is published by Publisher.

Book_Authors = Bookid   link authors to books

PUBLISHER = No FK

Book.copies = (bookid, branchid)

Each book copy has a book and books quantity at each branch.

Book_loans = (bookid, branchid, CardNo)

Which borrower borrowed which book from which branch.

Library_Branch = branchid every library link with branch.

BORROWER = CardNo   CardNo links borrower.

(4) Book: bookid → INT

Title → VARCHAR(100)

PublisherName → VARCHAR(100).

Book_Authors: bookid → INT

AuthorName → VARCHAR(100).

PUBLISHER: Name → VARCHAR(100)

Address → VARCHAR(100)

Phone → VARCHAR(50)

Book_COPIES: bookid → INT

branchid → INT

No-of-copies → INT

Book.LOANS: bookid → INT

branchid → INT

CardNo → INT

Dateout → DATE

DueDATE → DATE

Library_BRANCH: branchid → INT

branchName → VARCHAR(100)

Address → VARCHAR(100).

BORROWER: CardNo → INT

Name → varchar(50)

Address → VARCHAR(50)

Phone → VARCHAR(50).

(5). The relationship will be one to many because one author can write multiple books.

Bookid	Author
101	John
102	John
103	John
104	John

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Q5 1) Data indepence in DBMS means changes in database structure doesn't affect application program.
(1) Physical independence
(2) logical independence
example :- suppose student table is stored in heap file later file changed in indexed file
2) A teacher daily assigns lab task marks in excel sheet. 1- easy to use
2- No complex setup 3- low cost 4- suitable for small data.

(3) DELETE keyword remove the specific tuple, table remains
TRUNCATE removes all rows, TABLE Remains.
DROP delete entire table, Table deleated.

(4) Regrential integrity ensure foreign key values match primary key values in the regrential table.
It prevents invalid data, Maintains consistency and avoid orphan records.

(5) Yes forcing key can have Null.
The relationship is optional, the colum is not defined as NOT Null
Effect on regretial integrity:
Null doesnot violate regrential integrity, because Null means no relationship.